Activity 1 - Map the GHG System and Audit Criteria

Alignment: K1 | K2 | K8 | A1 | A8 | LO1 Duration: 45 minutes Objective: Translate ISO 14064 and related criteria into a process map and auditable requirements register.

Scenario

HarbourLight Electronics Pte Ltd manufactures high-efficiency LED panels in Tuas. Its 2025 GHG report was assembled by Sustainability from utility bills, procurement spreadsheets and supplier estimates. The company has ISO 14001 certification, but GHG responsibilities, data hand-offs and review approvals are not mapped. A major customer now requests a verification-ready organisational inventory and product-footprint evidence for the Lumina 400 LED panel.

Materials

- HarbourLight organisation and process facts
- ISO 14064 family and related-framework summaries
- Draft GHG report contents
- Process-map and criteria workbook

Questions

- Which ISO 14064 parts apply to each assertion?
- How does ISO 14001 support but not replace GHG criteria?
- Where does evidence enter, change and receive approval?
- Which report sections need a traceable owner and criterion?

Detailed procedure

1. Read the fact pattern and separate organisational-inventory, product-footprint and verification assertions.
2. List intended users, decisions, reporting period and desired assurance readiness for the audit.
3. Map the end-to-end GHG process from source activity through factor selection, calculation, review, approval and disclosure.
4. Identify roles and evidence hand-offs among Facilities, Finance, Procurement, Logistics, Product Engineering and Sustainability.
5. Build a criteria crosswalk for ISO 14064-1, relevant ISO 14001 controls, ISO 14064-3 concepts and declared GHG Protocol methods.
6. Flag criteria conflicts, gaps and items that require authorised standards text or programme clarification.
7. Create a document request list covering boundaries, source records, factors, calculations, uncertainty, approvals and claims.
8. Design a reporting index that links each disclosure to criterion, owner, source evidence and reviewer.
9. Peer-review the process map for missing stages and resolve one disagreement using evidence.

Required evidence

- GHG system process map
- Criteria crosswalk
- Document request list
- Reporting and evidence index

Acceptance check

Every assertion is assigned to declared criteria; the process map shows all major data and approval hand-offs; the reporting index links requirements to retrievable evidence.

Files

- Activity-01-Workbook.xlsx - editable working file
- Activity-01-Activity-Pack.pdf - printable scenario, questions, procedure and checklist

Use fictional, anonymised or authorised workplace information only.